Deploying Wazuh SIEM and Implementing File Integrity Monitoring (FIM) on Windows Server

Introduction

This documentation explains how to deploy a Wazuh Security Information and Event Management (SIEM) platform using an OVA file on VMware Fusion (macOS host), and how to configure File Integrity Monitoring (FIM) on a Windows Server endpoint. This walkthrough documents each step taken and provides checkpoints for review.

Lab Environment and Requirements

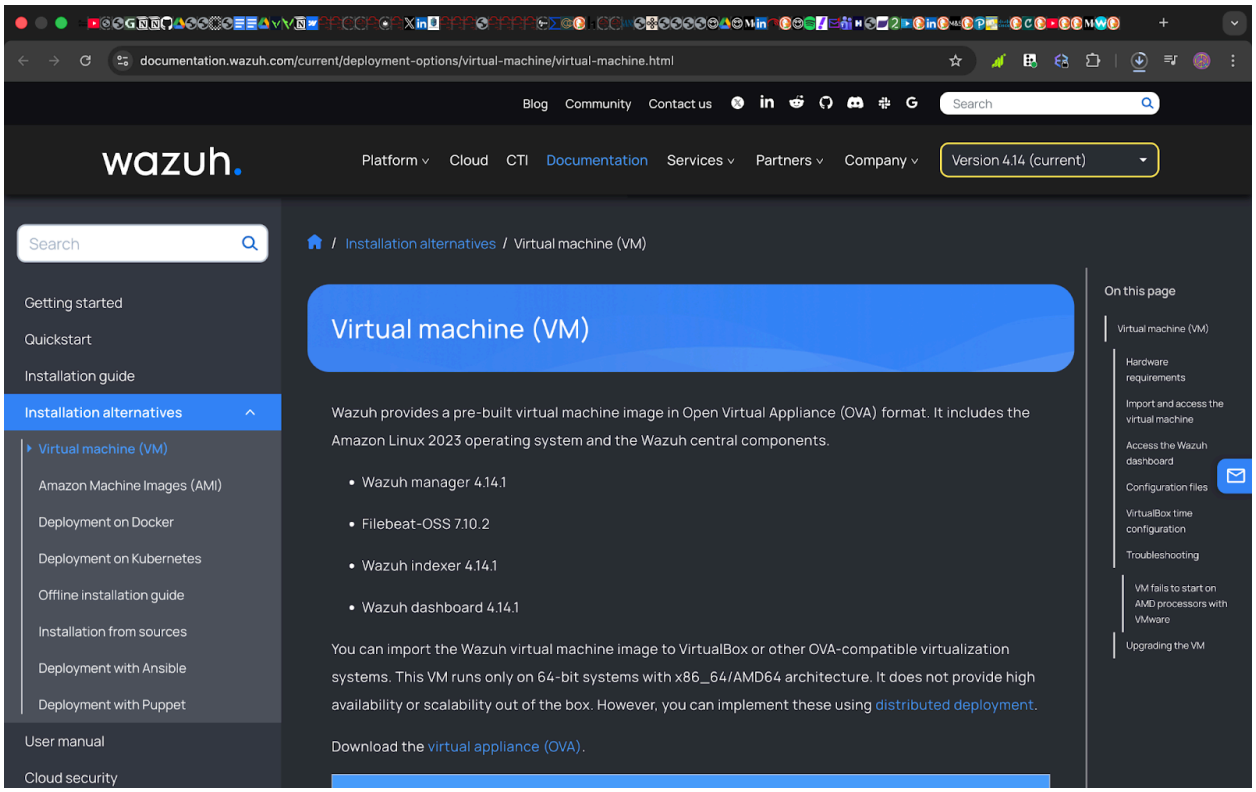
a. Host Machine: macOS (MacBook)
b. Virtualization Platform: VMware Fusion
c. SIEM Platform: Wazuh All-in-One (OVA deployment)
d. Endpoint 1: Windows Server 2019 / 2022
e. Endpoint 2: Kali Linux
f. Network Mode: NAT or Bridged

1. Deploying Wazuh Manager Using OVA

The Wazuh Manager was deployed using the official All-in-One OVA image. The OVA file was imported into VMware Fusion, allocated sufficient CPU and memory resources, and configured with appropriate network settings. After powering on the virtual machine, the Wazuh services initialized automatically.

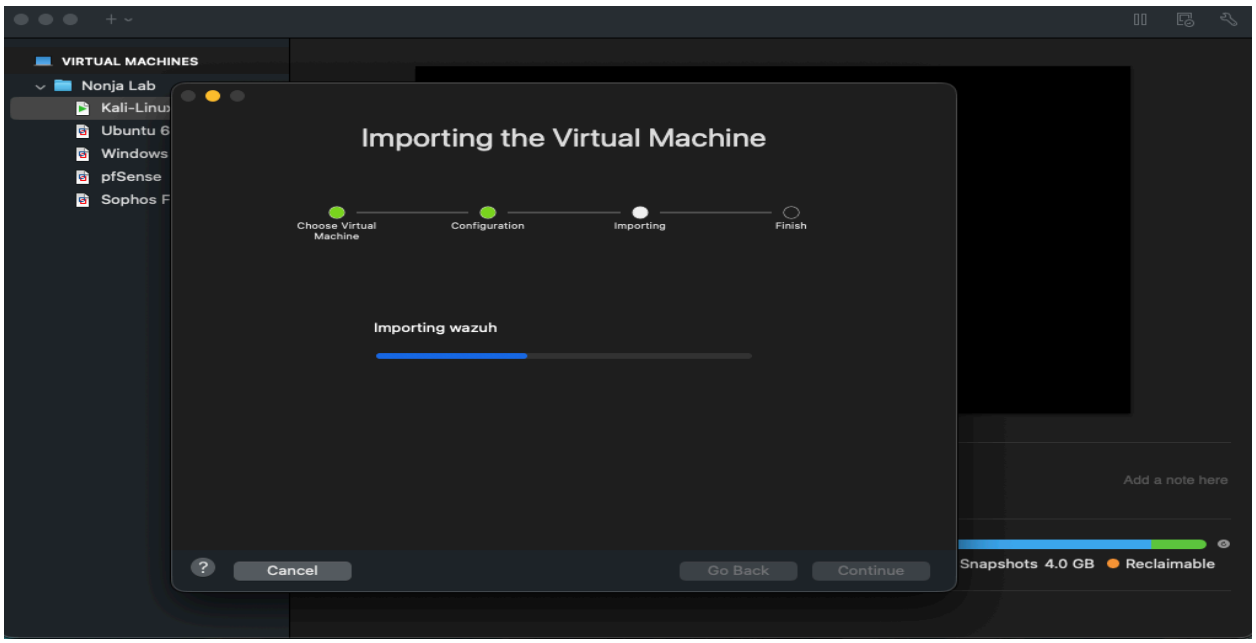
1.1 Downloading the Wazuh OVA

1. Navigate to the official Wazuh Documentation and Downloads page.
2. Download the Wazuh All-in-One OVA file suitable for virtualized environments.



1.2 Importing the OVA into VMware

1. Launch VMware Workstation / VMware Fusion.
2. From the menu, select File → Import.
3. Browse to the downloaded .ova file and select it.
4. Assign a name to the virtual machine (e.g., Wazuh) and choose a preferred storage location.
5. Click Save to begin the deployment process.

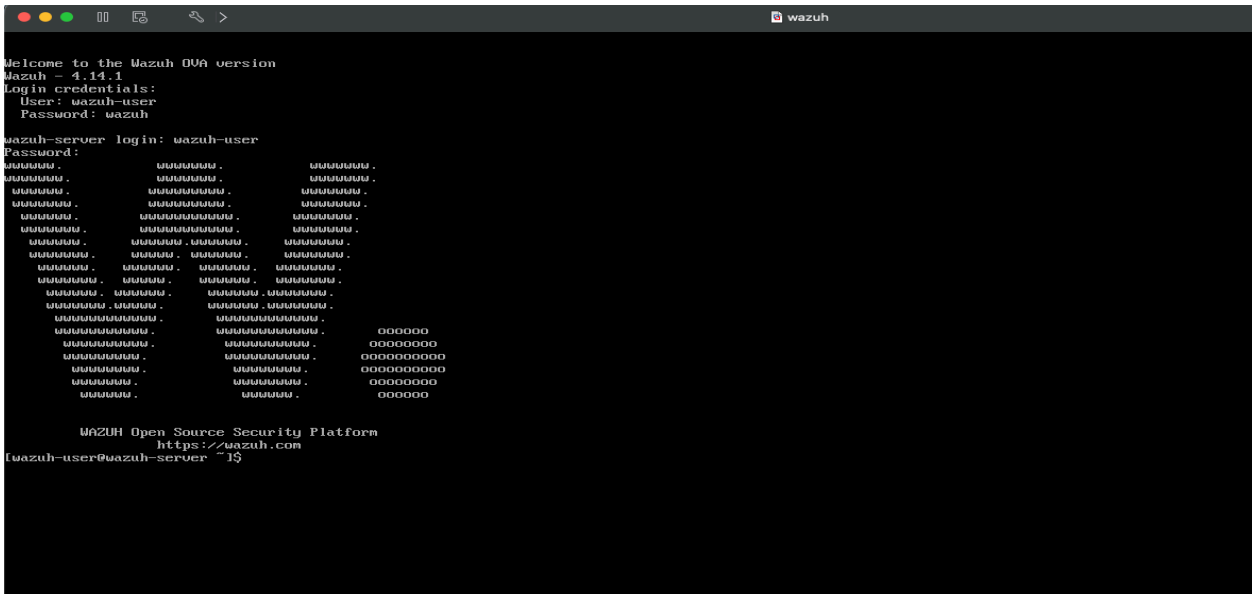


6. After the import completes, power on the virtual machine.

1.3 Initial System Configuration and Access

- Once the virtual machine boots, log in through the VMware console using the default credentials provided on the Wazuh download page

  (commonly: wazuh-user / wazuh, depending on the version).



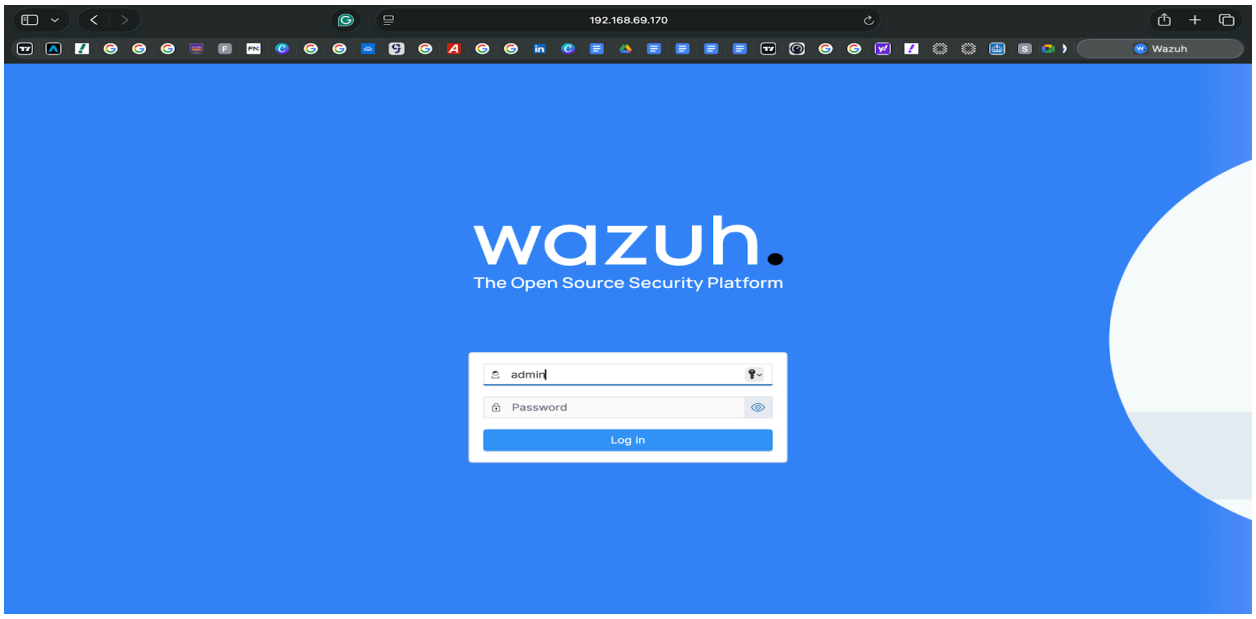
- After successful login, obtain the system's IP address by running: ***ifconfig***
- Note the assigned IP address, which is typically provided via DHCP from VMware (for example, `192.168.69.170`).

1.4 Verify Wazuh Manager Services

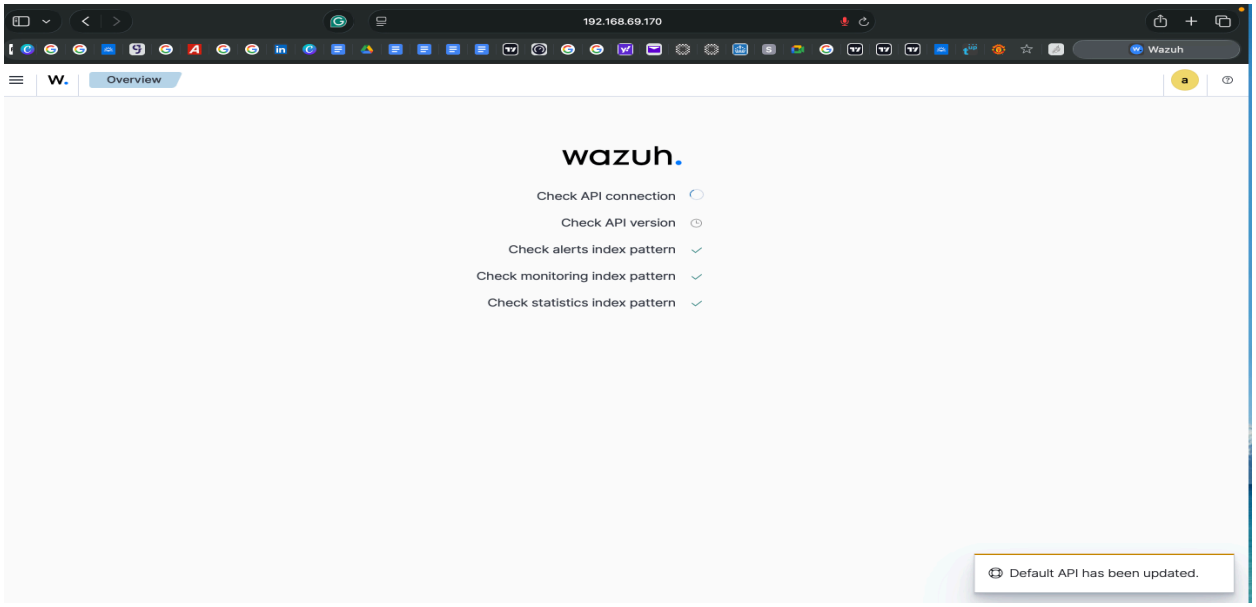
→ sudo systemctl status wazuh-manager
→ sudo systemctl status wazuh-indexer
→ sudo systemctl status wazuh-dashboard

1.5 Accessing the Wazuh Dashboard

- From the host machine, open a web browser and navigate to: `https://<WAZUH_IP>`
- Accept the browser security warning caused by the self-signed SSL certificate.
- Log in to the Wazuh web dashboard using the default credentials:
  - Username: admin
  - Password: admin



Once the Wazuh virtual machine completed booting, the assigned IP address was used to access the Wazuh dashboard via a web browser on the host system. Default administrative credentials were used for initial login.

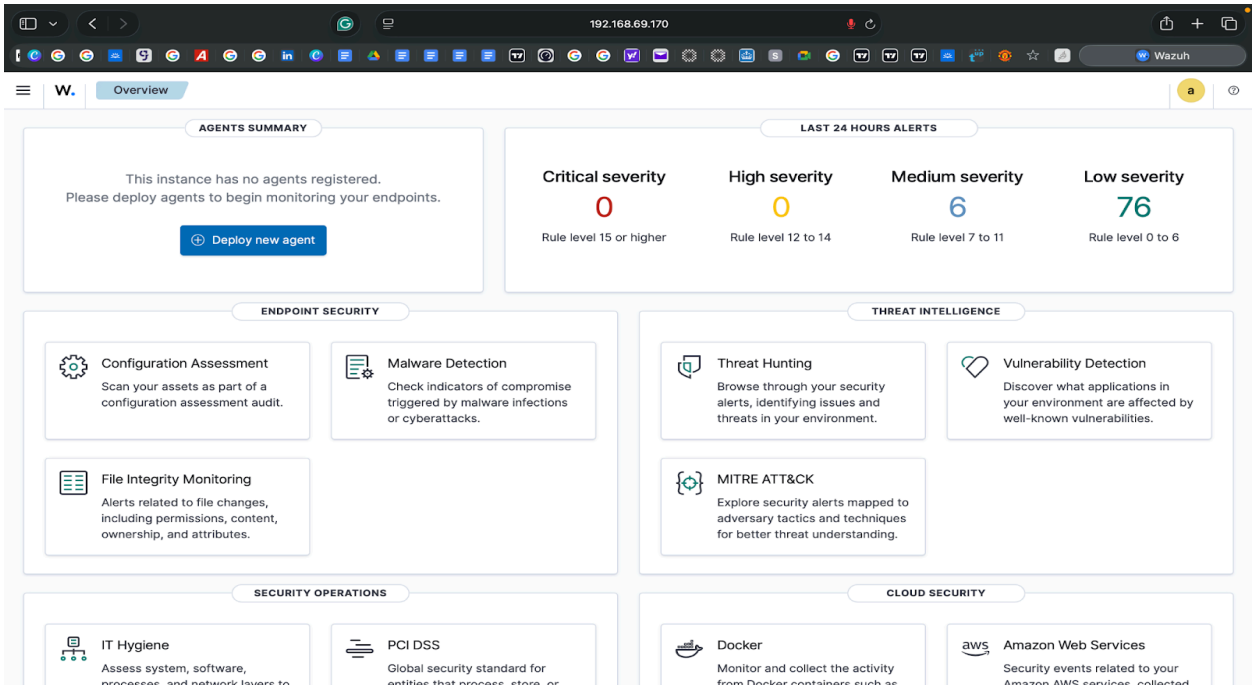


2. Installing Wazuh Agents on Endpoints

In this phase, Wazuh agents are deployed to the endpoint systems using the Wazuh Dashboard installation wizard. This approach ensures consistency and minimizes errors during agent configuration.

2.1 Deploying the Wazuh Agent on Windows Server 2022

1. Login to the **Wazuh Dashboard** from the host machine's web browser.
2. Click **Deploy new agent,** available on the overview tab.

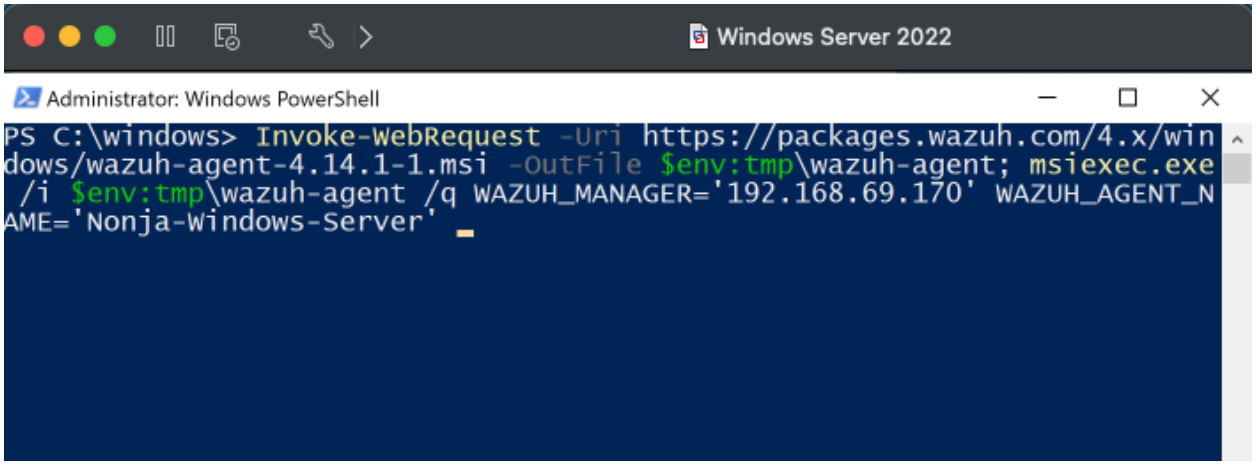


3. **Operating System Selection:** Choose **Windows**.
4. **Server Address:** Enter the Wazuh Manager IP address (192.168.69.170).
5. **Agent name: "**Give your agents name for easy identification"
6. **Agent Group:**
   Leave the default group selection unchanged
7. Command Execution:
   Copy the installation command generated by the dashboard

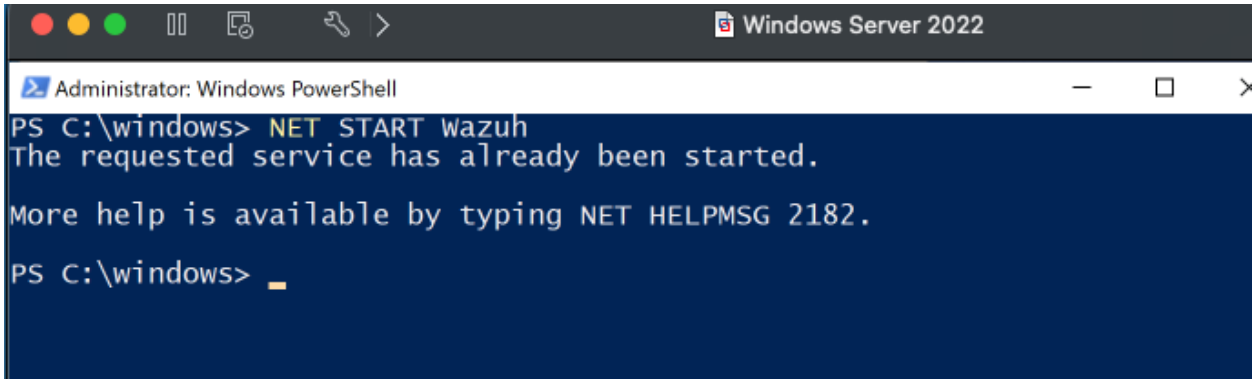
Run the following commands to download and install the agent:

```
Invoke-WebRequest -Uri https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.1-1.msi -OutFile $env:tmp\wazuh-agent; msiexec.exe /i $env:tmp\wazuh-agent /q
WAZUH_MANAGER='192.168.69.170' WAZUH_AGENT_NAME='Nonja-Windows-Server'
```

8. On the Windows Server 2022 VM:
   - Open PowerShell as Administrator.
   - Paste and execute the generated command.



9. **Start the Wazuh agent service:** NET START Wazuh



2.2 Deploying the Wazuh Agent on Kali Linux

1. Return to the **Add agent** page on the Wazuh Dashboard.

2. Log in to the Wazuh Dashboard from the host machine's web browser.
3. **Operating System Selection:**
   Select Debian/Ubuntu, as Kali Linux is Debian-based.

4. **System Architecture:**
   Choose amd64.

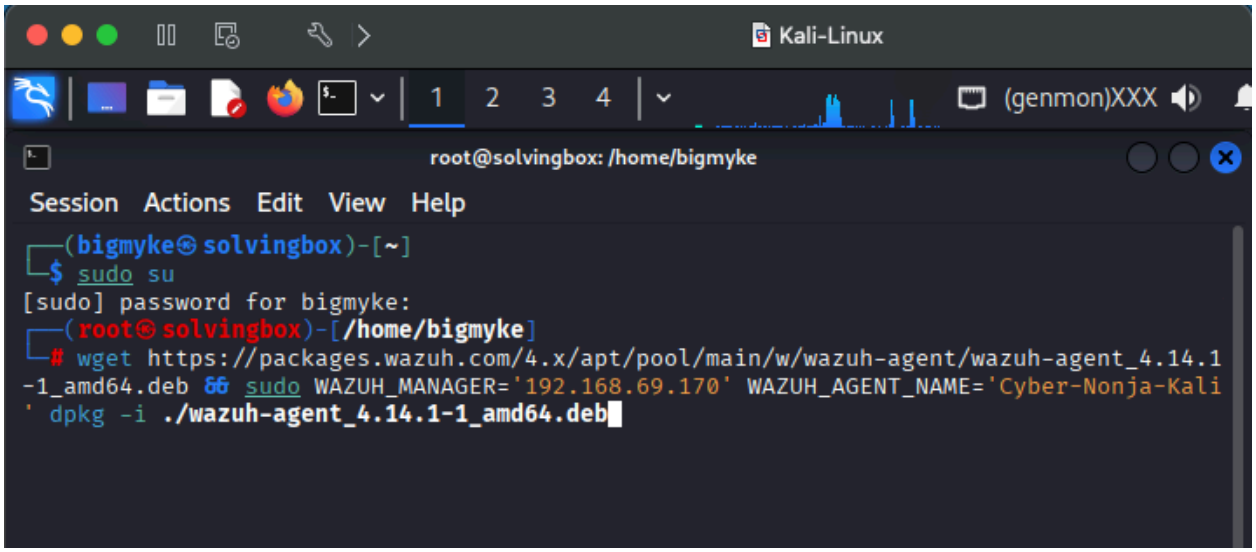
5. **Server Address:** Enter the Wazuh Manager IP address (192.168.69.170).

6. **Agent name:** **“**Give your agents name for easy identification”

7. **Agent Group:**
   Leave the default group selection unchanged

8. **Command Execution:**
   Copy the installation command generated by the dashboard

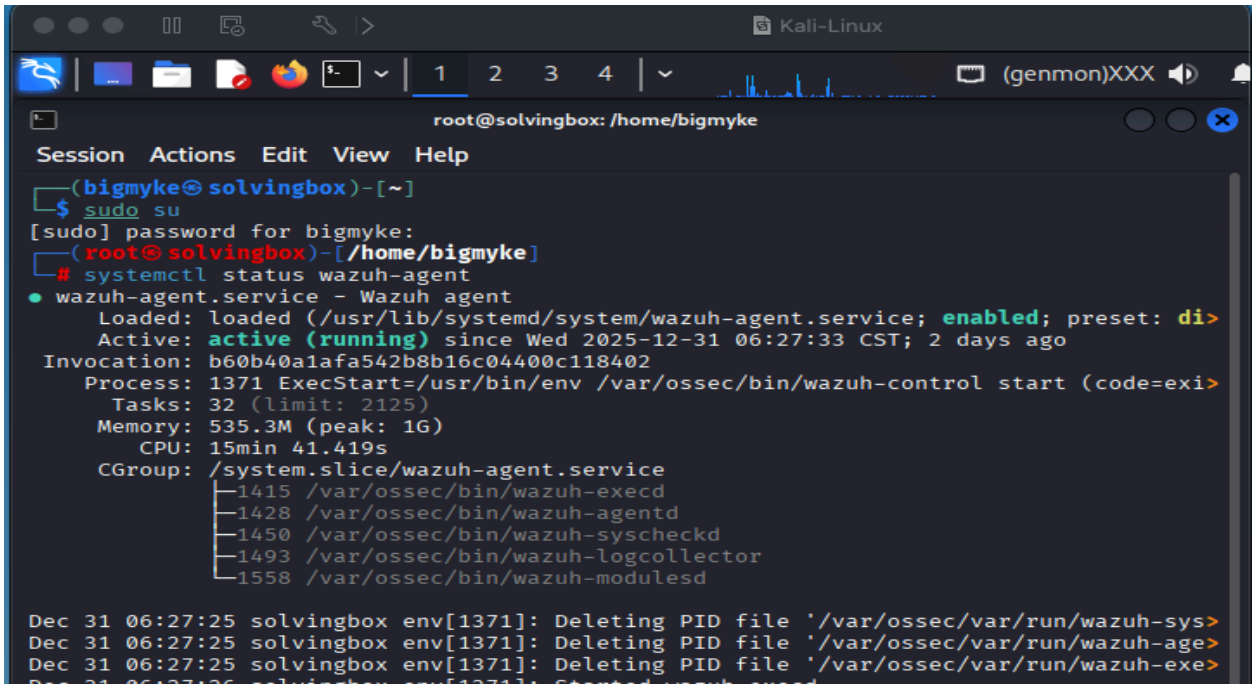
9. On the Kali VM, open a terminal and paste the command using elevated privileges(sudo), then execute it.



10. Enable and **start** the Wazuh agent service:

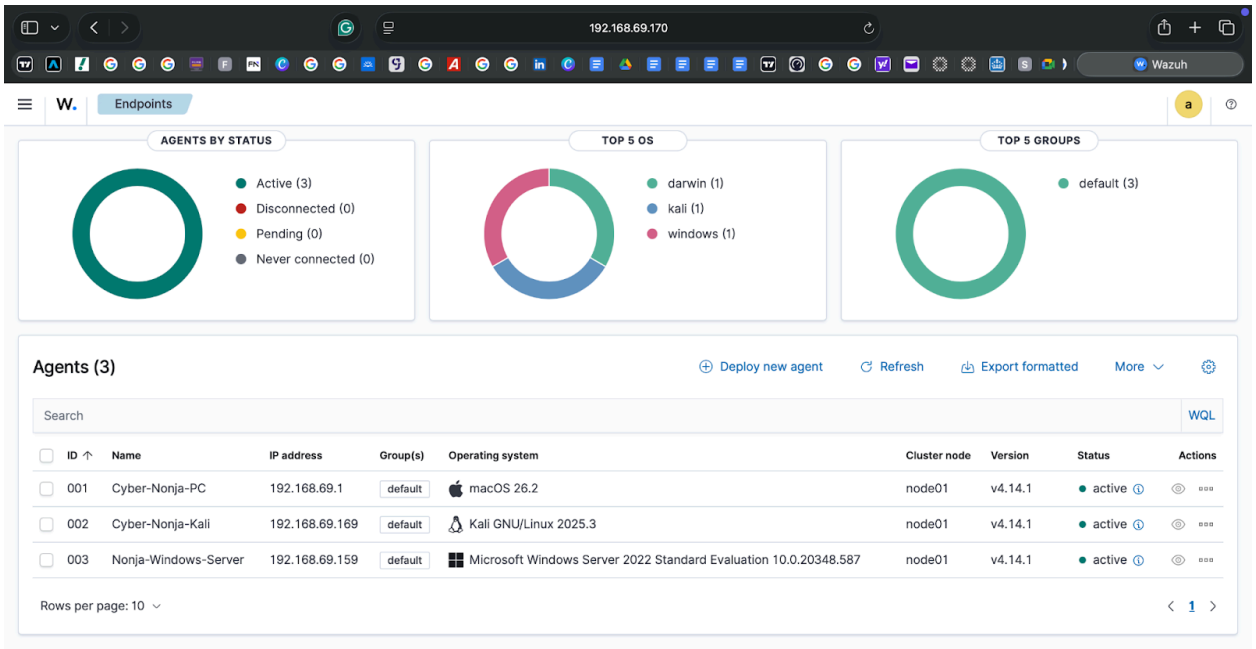
5 **Start the agent:**

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```



2.3 Agent Connectivity Verification

1. Navigate to Menu → Agents in the Wazuh Dashboard.
2. Confirm that both Windows Server 2022, Kali Linux and any other agents deployed appear with an **Active status**.



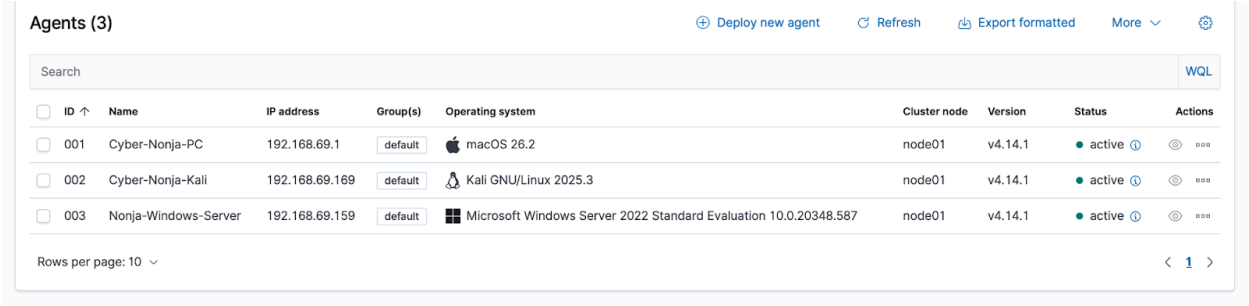
The Wazuh agent was installed on a Windows Server system to serve as the monitored endpoint and also on Kali for simulation purposes. During installation, the agents was registered with the Wazuh Manager using its IP address. Successful registration was confirmed from the Wazuh dashboard.

3. File Integrity Monitoring (FIM)

File Integrity Monitoring is configured to detect unauthorized changes to critical files and directories. This capability directly supports **PCI DSS Requirement 11.5.**

3.1 Configuring FIM on Windows Server 2022

1. Confirm Windows Agent is Connected:

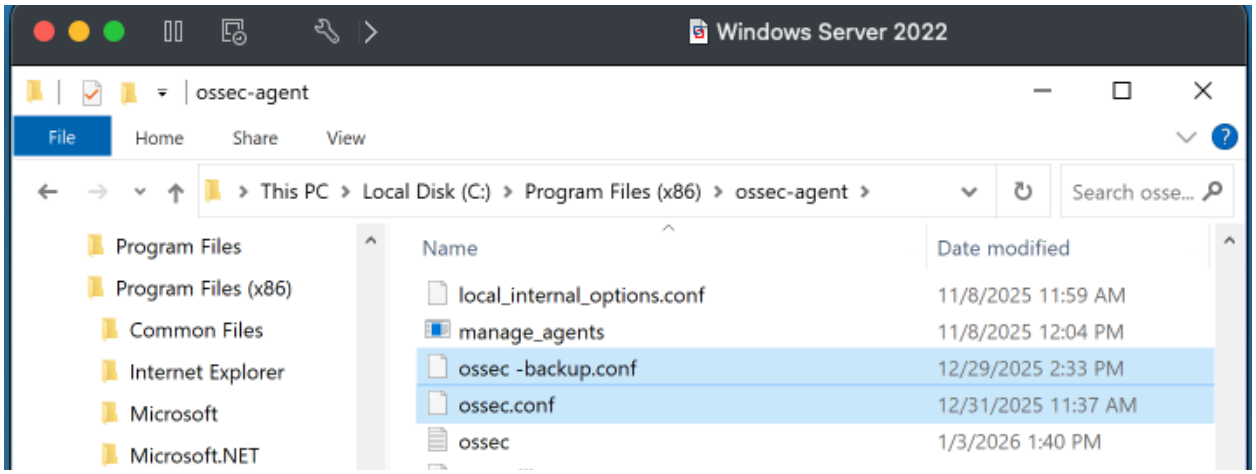


✔ Windows agent = **Active**

2. Locate **Wazuh Agent Configuration** on WIndows Server:

   C:\Program Files (x86)\ossec-agent\ossec.conf

   This PC → C: (Local Disk) → Program Files (x86) → ossec-agent
   → Backup Existing Configuration (Important):



   Open with:
   - **Notepad++** (recommended)
   - Or Notepad (Run as Administrator)

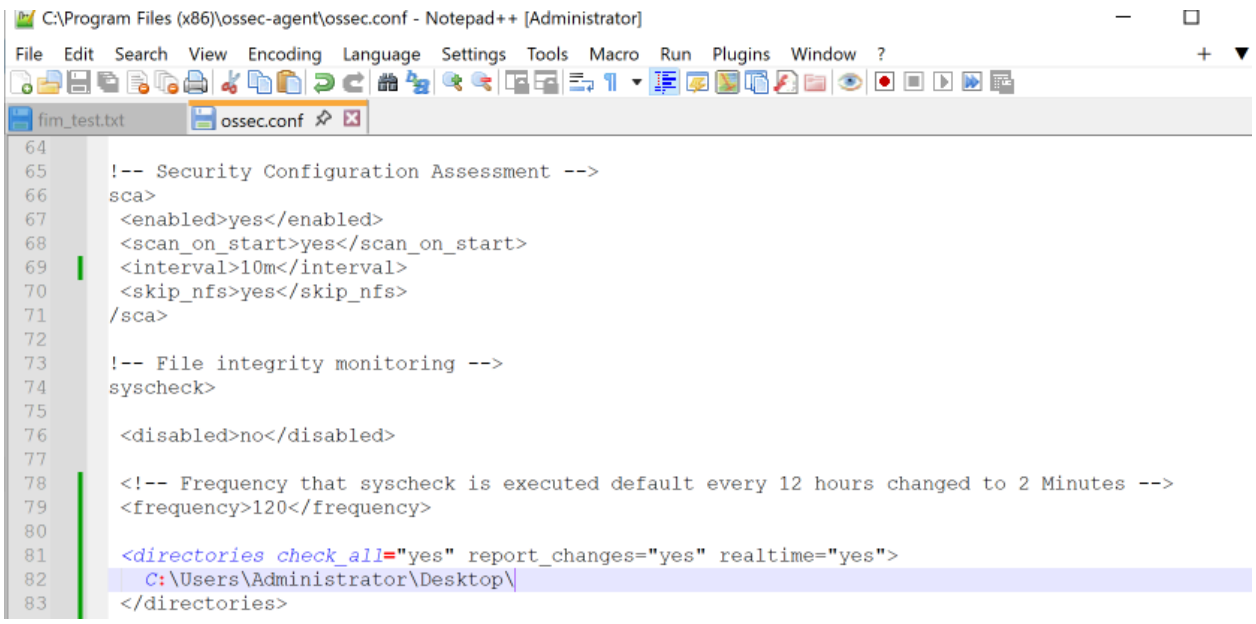
3. Inside the ossec.conf file:

   → Locate the <syscheck> configuration block.

   → Add the following directory monitoring rule:

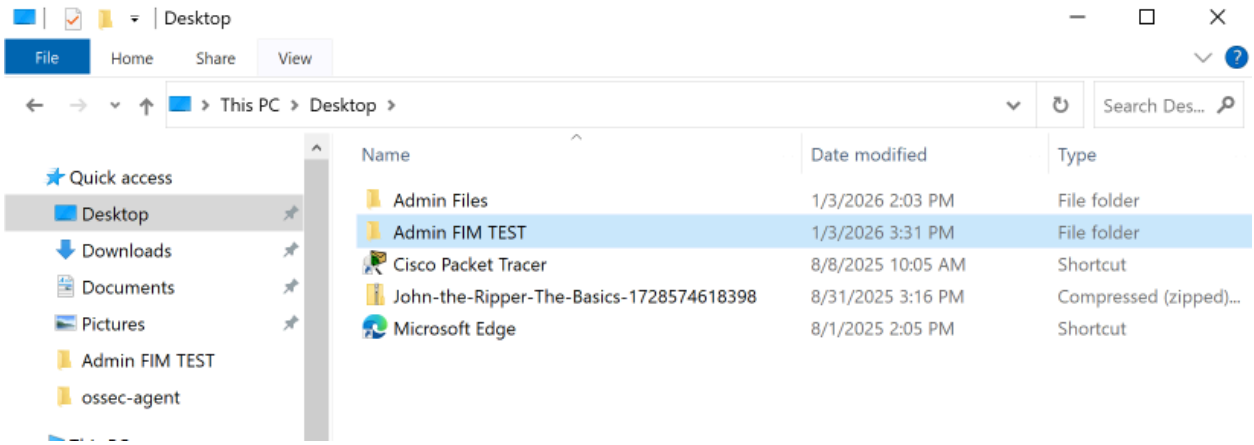
```
<directories check_all="yes" report_changes="yes" realtime="yes">
  C:\Users\Administrator\Desktop\
</directories>
```

   → Also change the frequency to 120 sec (2 minutes) and interval to 10minutes to enable faster detection of any changes.

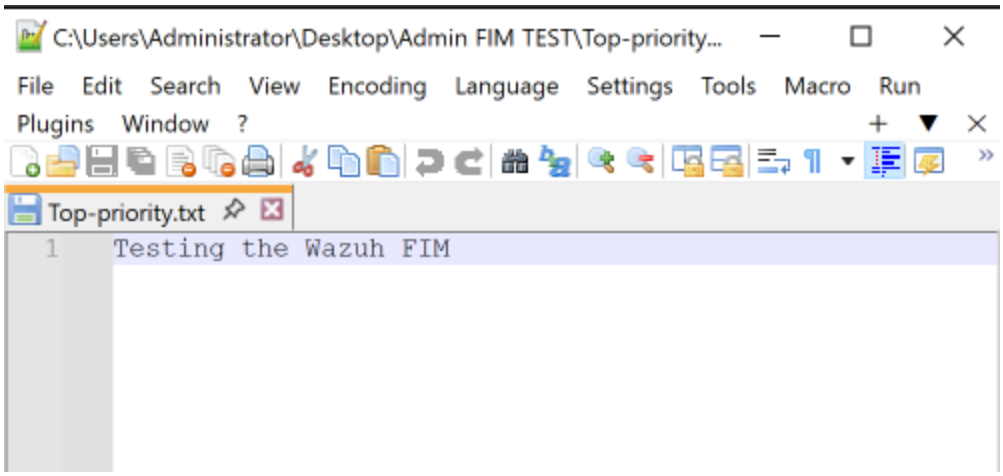


Note: `report_changes` and `realtime` enable faster and more detailed change detection.

4. Restart the Wazuh agent service.
5. Create a new folder on the Desktop of the Windows server and name it (*for testing purpose*).



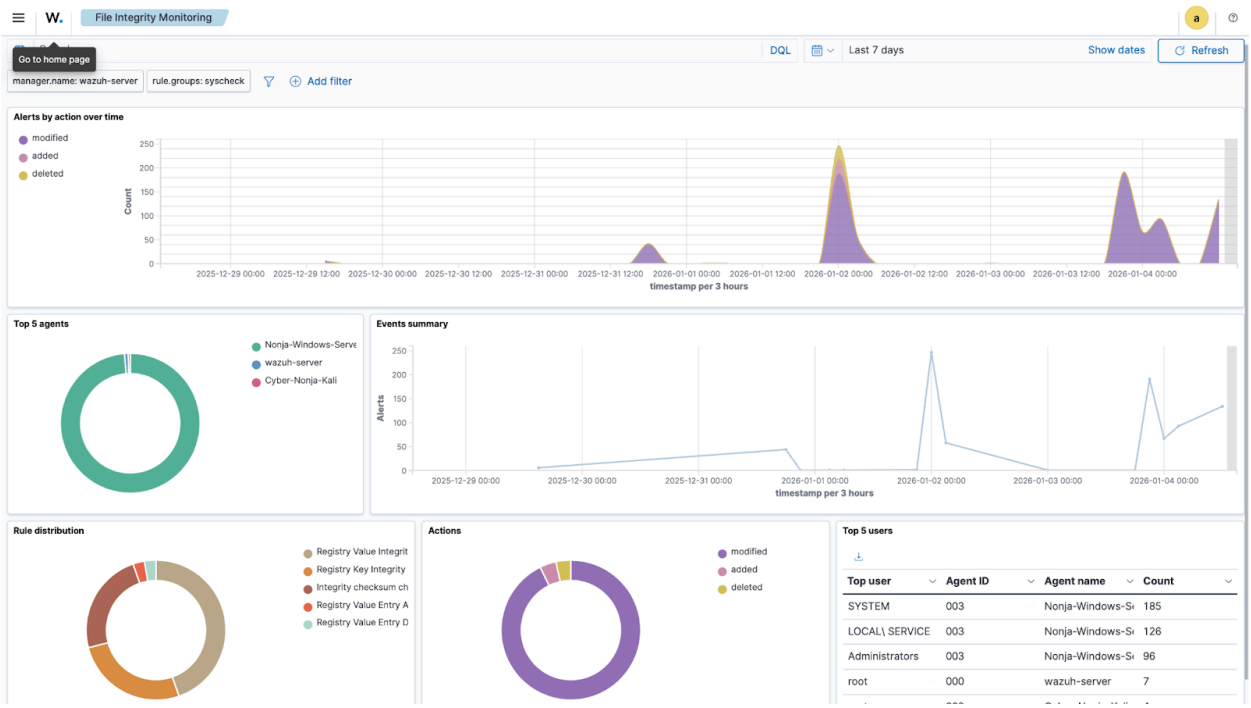
6. Open the folder and create a text file and name it (for testing **purposes.txt)**.
7. Open the file, add random content, and save it.



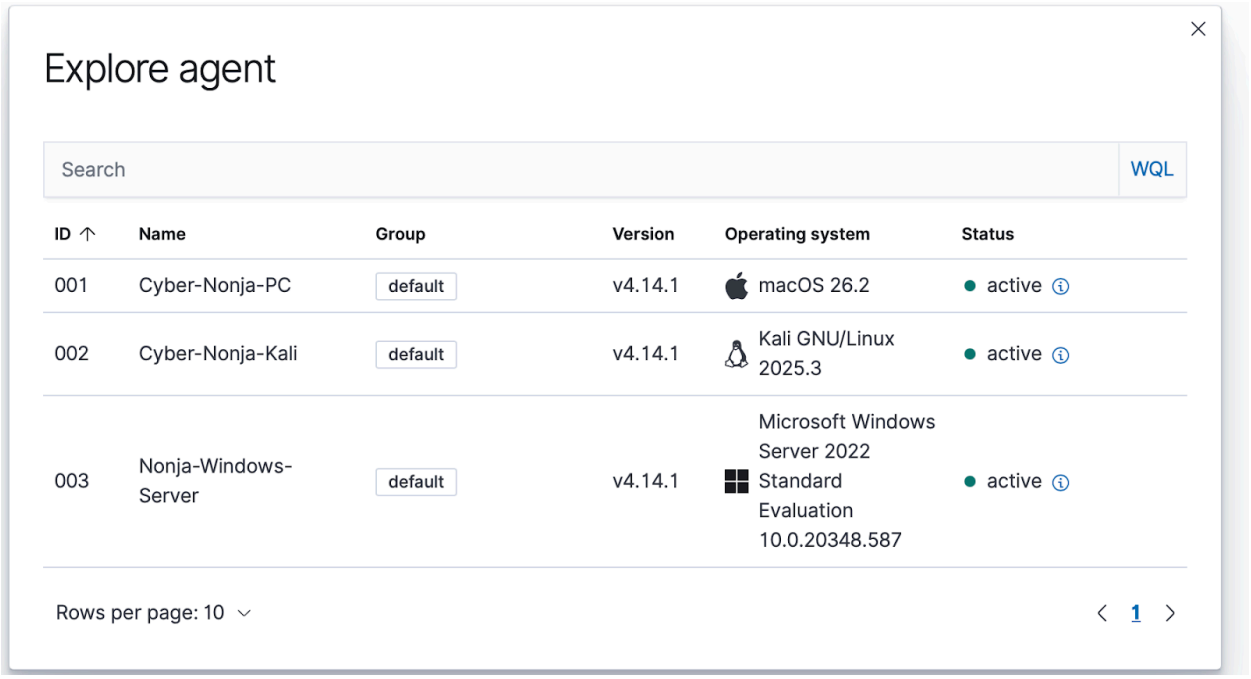
File Integrity Monitoring was configured by editing the Wazuh agent configuration file (ossec.conf) on the Windows Server endpoint. Critical system directories paths were added to the monitoring scope to detect unauthorized changes.

3.2 Verifying FIM Events in the Dashboard

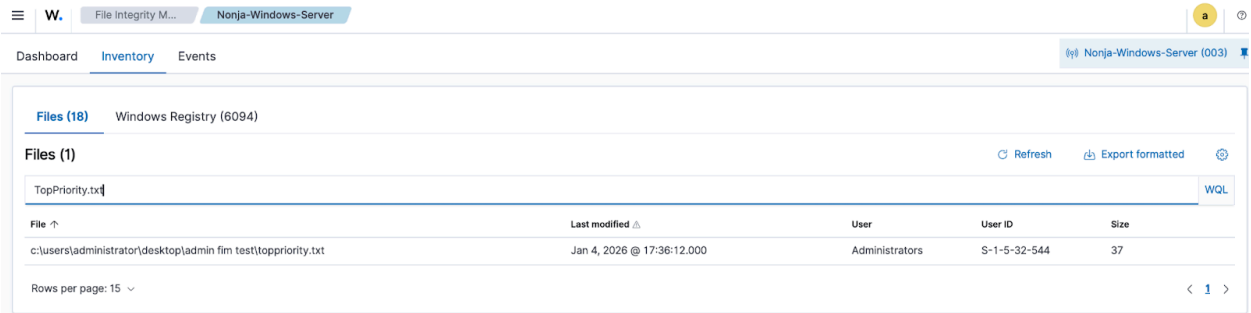
1. Navigate to **Menu** → **Endpoint Security** → **File Integrity Monitoring**.
2. Ensure the **Show dates** time range covers the duration of the lab **[Last 24hours or Last 7 days]**.



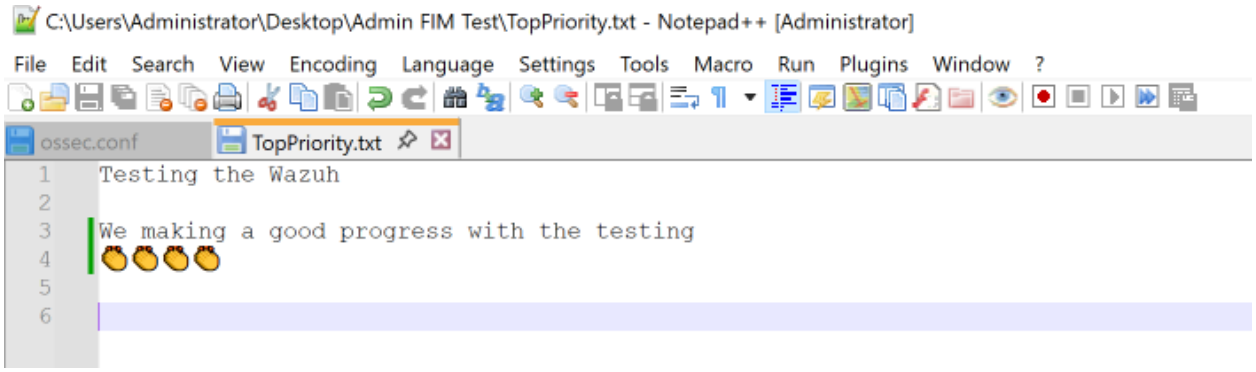
3. Switch to the **Inventory** tab and select the Windows Server agent.



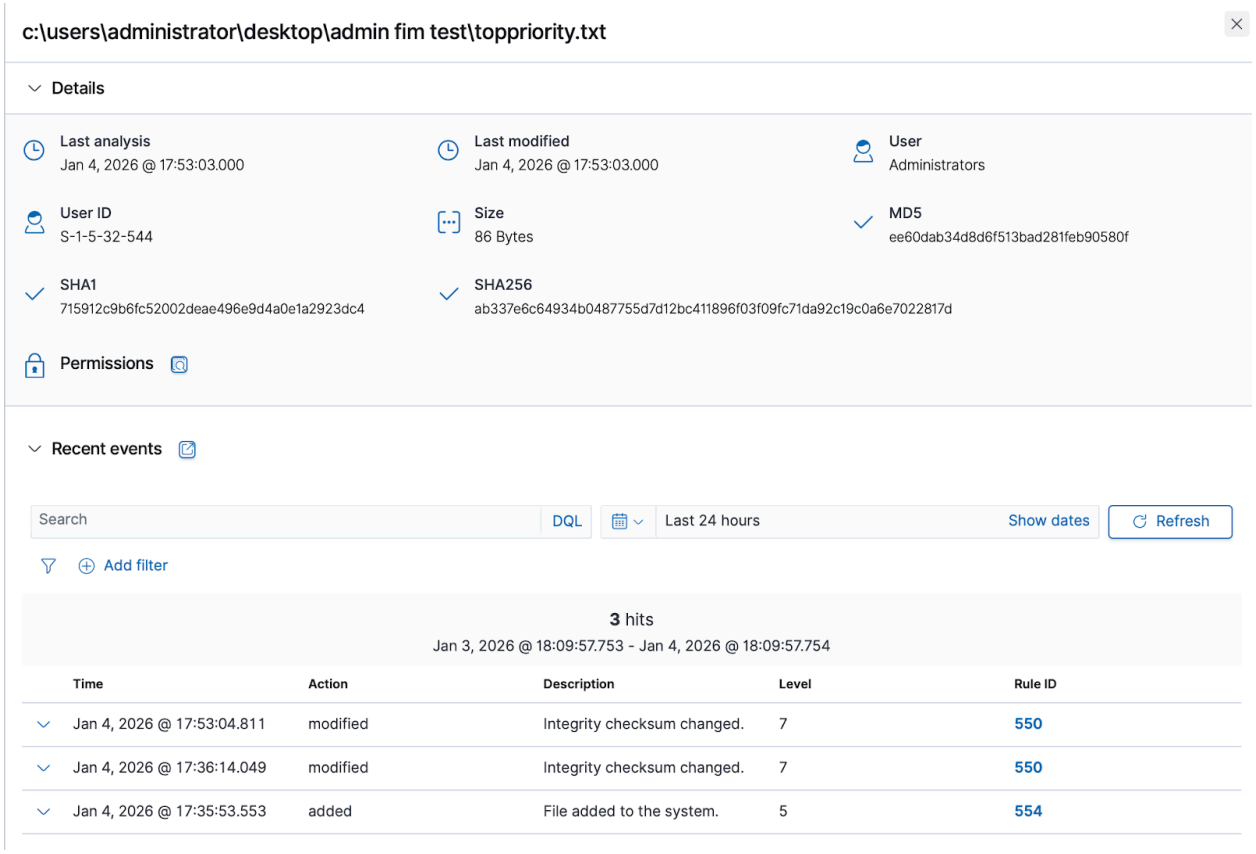
4. Search for the `file you created` {**`Toppriority.txt`**} and confirm it appears.



5. Modify the file by adding new content and save it.

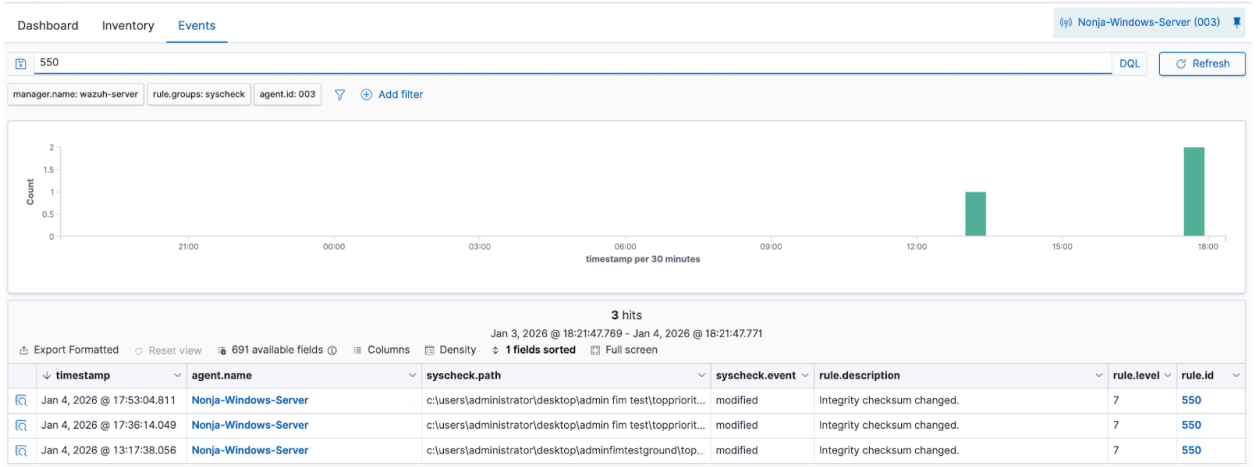


6. Return to the Inventory tab and open the file details.
7. Verify an event labeled **“Integrity checksum changed”** is present.

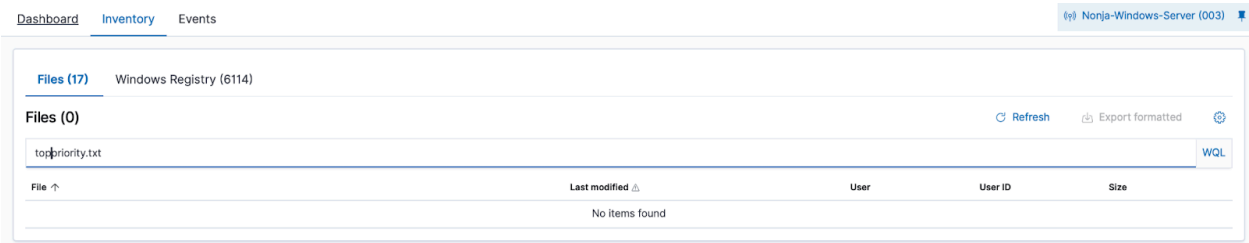


8. Navigate to the **Events** tab and search for:

   `rule.id: 550`



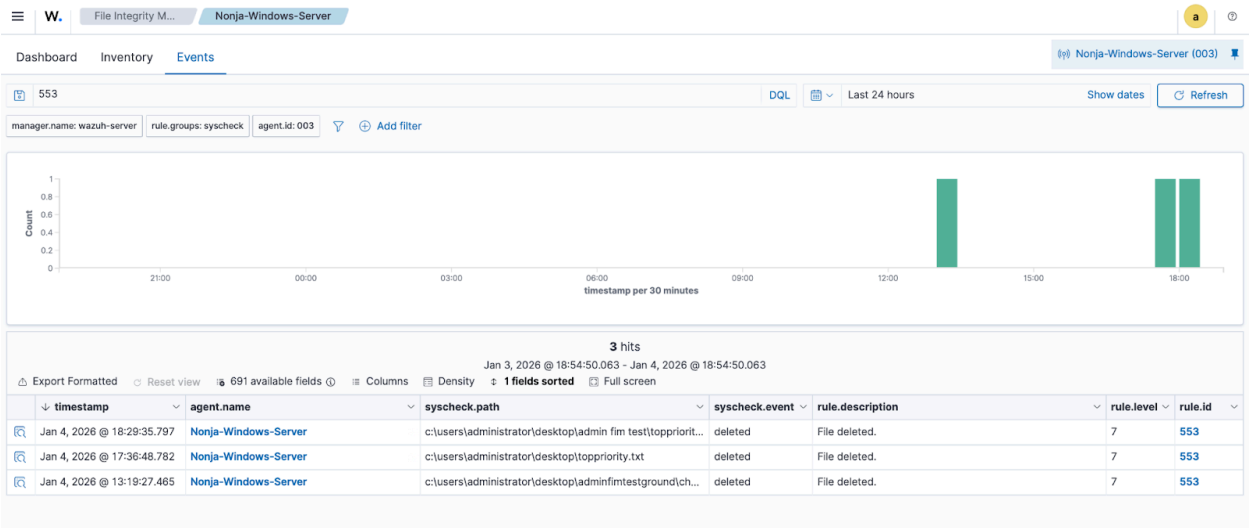
9. Delete the monitored file from the **Admin FIM Test** folder.
10. Confirm the file no longer appears in Inventory results.



11. Navigate to the Events tab and search:

    `rule.id: 553`

12. Verify the file deletion event is logged.



4. Monitoring Security Events and Alerts

1. Navigate to Menu → Threat Intelligence → Threat Hunting.
2. **Summary:**

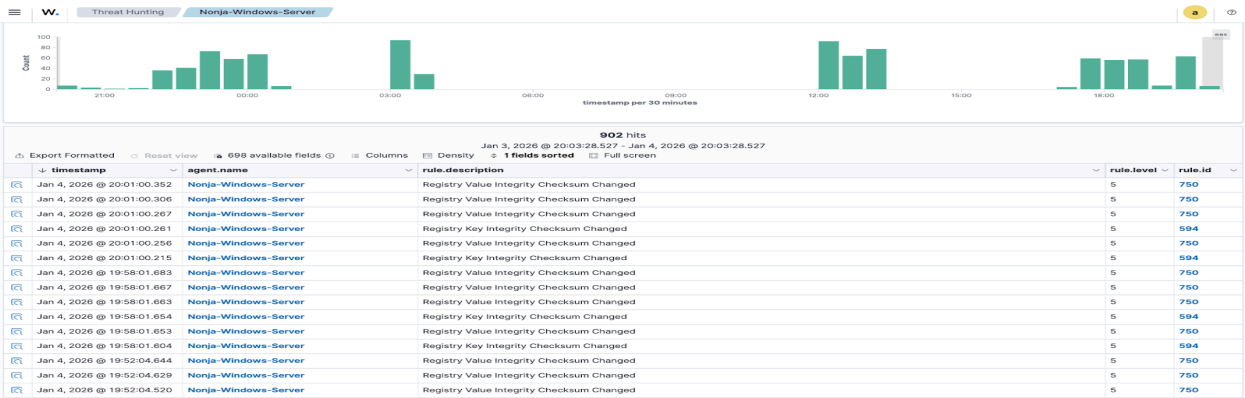
   → **Threat Hunting Dashboard:**

   The dashboard provides a high-level overview of security activity on the Nonja-Windows-Server endpoint. A total of 902 events were recorded, with no high-severity (Level 12+) alerts. Most activity is related to File Integrity Monitoring and registry integrity checks, confirming that monitoring is active and functioning correctly.

Threat Hunting Dashboard

→ **Threat Hunting Events:**

The Events view displays detailed logs for all detected activities. The majority of events are registry value and registry key integrity changes (Rule IDs 750 and 594), all logged at rule level 5. These events were generated as a result of monitored changes on the Windows Server endpoint.



Threat Hunting Events

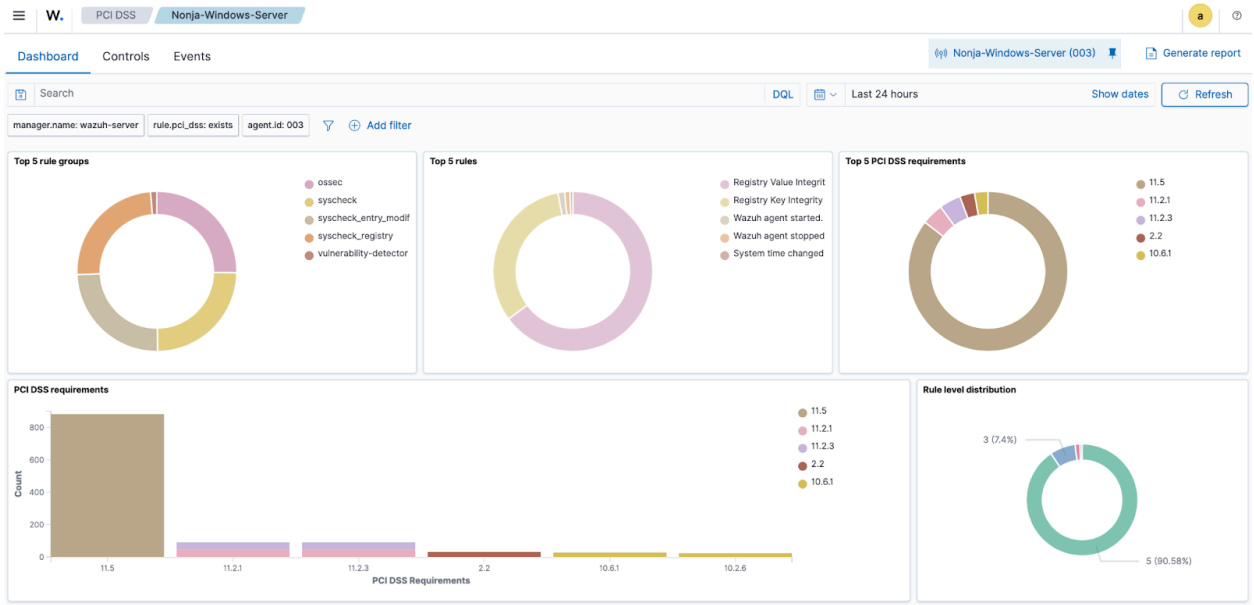
5. GDPR and PCI DSS Compliance Monitoring

Wazuh automatically links security events to compliance requirements when corresponding rules apply.

5.1 PCI DSS Compliance Review

1. Navigate to **Menu** → **Security Operations** → **PCI DSS**.

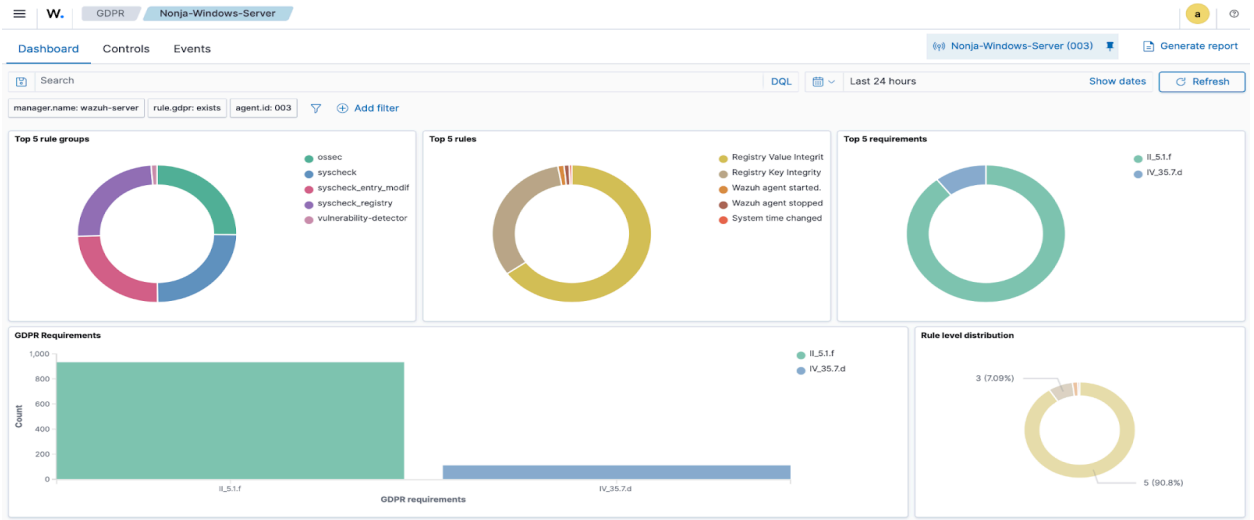
2. Review the compliance dashboard.



The PCI DSS dashboard shows that most security events for the **Nonja-Windows-Server** agent map to **Requirement 11.5**, confirming that **file and registry integrity monitoring** is functioning correctly. A smaller number of events relate to other requirements such as **11.2** and **10.6**, with most alerts classified as **low-to-medium severity**.

5.2 GDPR Compliance Review

1. Navigate to **Menu** → **Security Operations** → **GDPR**.
2. Review mapped GDPR articles and alerts.



The Wazuh GDPR dashboard provides an overview of security events that are automatically mapped to relevant GDPR requirements based on triggered detection rules.

In this project, file integrity and system change events from the Windows Server were successfully correlated with applicable GDPR articles, allowing visibility into compliance-related activities and alert severity.

This confirms that Wazuh can effectively support GDPR compliance monitoring through automated rule-based event classification.

6. Conclusion

This project successfully demonstrated the deployment of a Wazuh Manager using an OVA on VMware Fusion and the integration of a Windows Server endpoint for File Integrity Monitoring (FIM). Key security events, including file creation, modification, and deletion, were accurately detected and visualised in the Wazuh Dashboard.

The implementation confirmed effective agent communication, proper FIM configuration, and successful correlation of security events with compliance requirements such as GDPR. Overall, this setup validates Wazuh as a reliable open-source security monitoring and compliance solution suitable for enterprise environments.